

Part: **4.9 – COMPOSITE PARTS**
Chapter: **4920**
Title: **COMPOSITE DAMAGE THREATS**
Revision: **0.4**
Date: **23/01/2024**

1 INTRODUCTION

The purpose of this chapter is to assess the damage threats to be considered in structural justification of damage tolerant airframe composite parts (not exposed to exceptional damage risks), to be compliant with CS-25 regulations.

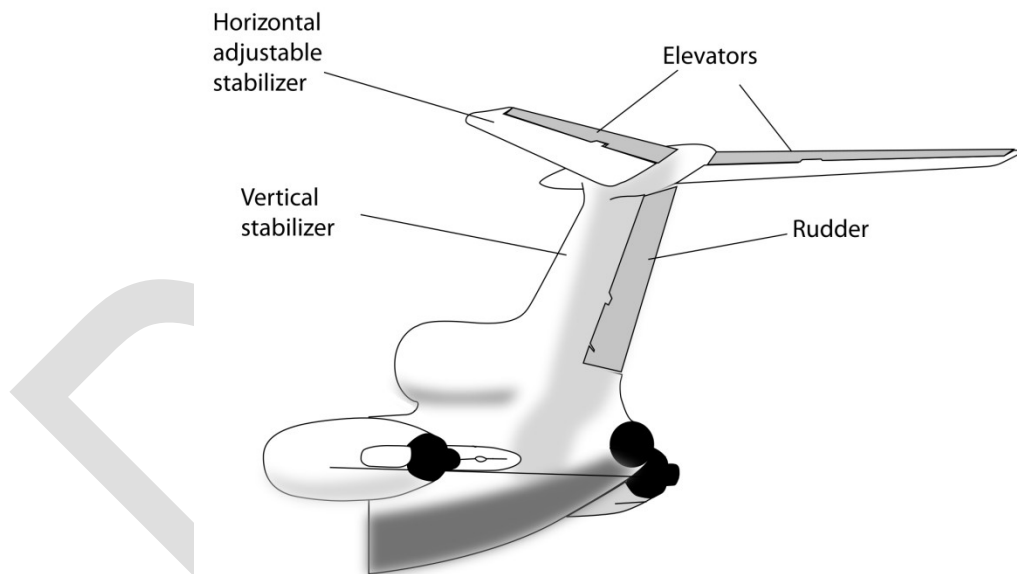


Figure 1. General arrangement of a T-tail (image taken from the Wikipedia: T-tail of the Tu-154)

The scope of this chapter intentionally limits actual discussion to the “typical” impact threats, avoiding parts with particular exposures (eg. fuselages at cargo doors, landing gears or propeller sections, or root zone of fuselage-mounted tail planes).

The following texts are extracts of AMC 20-29 (or AC 20-107B) regulations:

§7.f. “It should be shown that impact damage that can be likely expected from manufacturing and service, but not more than the established threshold of detectability for the selected inspection procedure, will not reduce the structural strength below ultimate load capability.”

8.a. “(1) Damage tolerance evaluation starts with identification of structure whose failure would reduce the structural integrity of the aircraft. A damage threat assessment must be performed for the structure to determine possible locations, types, and sizes of damage considering fatigue, environmental effects, intrinsic flaws, and foreign object impact or other accidental damage (including discrete source) that may occur during manufacture, operation or maintenance.”

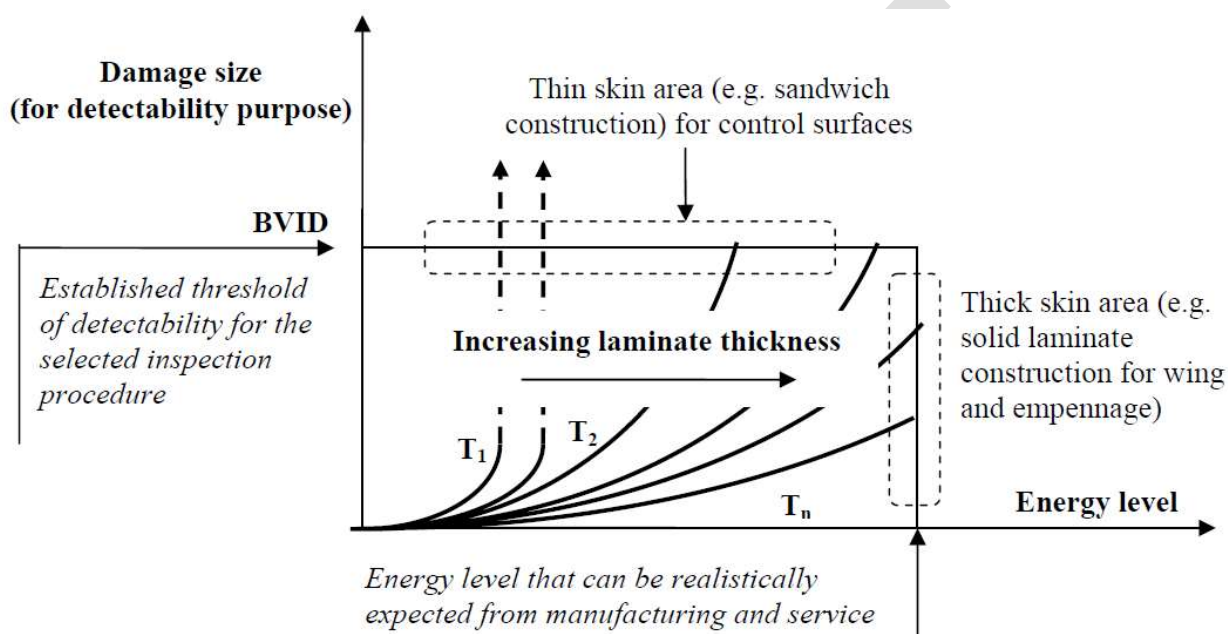


Figure 2. BVID limited by likeliness of occurrence (energy cut-off) or by detectability

Dealing with environmental conditions is accounted in qualifications processes of materials and fabrication, as well as in design values derivation. Detection and action on manufacturing defects are addressed by quality controls (including NDT inspections after curing/bonding structural composites) and adequate tuning of material design values to cover allowable defects. Fatigue damage is part itself of the damage tolerance assessment of the structure (see chapter 4930 “**DAMAGE TOLERANCE JUSTIFICATION AND DAMAGE TYPES**”). This report shall be focused in accidental damage, and foreign object impact damage in particular.

2 DAMAGE DETECTABILITY

Among the potential sources of damage, accidental damage is the most likely to reduce the strength capability of a composite structure. It is recognized that low velocity impacts, in particular those caused by blunt objects, can lead to large static strength reductions in compressively loaded structures without a visible indication of the damage.

Regarding accidental impact damage detection, visual inspection is the only one achievable within a reasonable cost. This is the reason why it is the sole option envisaged by the aircraft manufacturers

and operators, with the exception of the NDT inspections (typically ultrasonic inspections) done in composite parts after curing and bonding processes in manufacturing, prior to assembly processes.

Sensitivity of NDT inspection is clearly superior to visual inspections. Limits of allowable manufacturing defects, different than accidental damages, are typically set at levels criticality below the BVID and thus controlled and conservatively covered by the justification approach to low velocity impacts. No further discussion shall be included here regarding NDT inspections.

2.1 THE BVID THRESHOLD

The BVID (Barely Visible Impact Damage) defines the threshold of detectability, during scheduled or directed field inspection, for a damage caused by a low velocity impact, which the part could possibly suffer (with probability of occurrence above 10^{-5} /FH, i.e. the same of Limit Load event). Applicable damage metrics were built on the abovementioned conclusion of criticality for the case of impacts of blunt objects (including the normalization of the impactor in validation tests, **hemispherical of 16mm diameter within Airbus**).

a) Dent depth

Considering the dent depth as a damage metric is a widespread usage. Within Airbus the BVID thresholds actually set depend on the kind of inspection procedure. **For transverse impact these values are:**

- ✓ DVI: 0.3 mm after relaxation (*) (Ref. 11), or 1 mm after impact
- ✓ GVI: 1.3 mm after relaxation (*) (**)



Figure 3. Dents and holes after impacts of different energy levels in a panel

(*) After material ageing and relaxation. Typically initial dent depth is between 1.5 – 3 times the depth after relaxation. Impactors used for these studies are spherical, with diameters between 0.5 in. and 1.0 in., being the standard within Airbus a steel spherical impactor of 16mm diameter

(**) The dent depth after impact which will retain a dent depth after relaxation of 1.3mm is around 2.0-2.5 mm for CFRP with epoxy resins according to Airbus DS tests experience, but shall depend of the specific material system.

For edge impact these values are (Ref. 14 and Ref. 17):

- ✓ DVI: 0.3 mm after relaxation, or 0.6 mm after impact
- ✓ GVI: 1.3 mm after relaxation

b) Length of edge crack

This metric is particularly useful for impacts near to composite laminate edges. The BVID threshold considered within Airbus is 30 mm length in a Detailed Visual Inspection (DET/DVI) and 70 mm length in an General Visual Inspection (GVI) (Ref. 14) (*)



Figure 4. Edge crack

(*) BVID – DET is the threshold used by Airbus to define their allowable values for the residual strength after top edge impact failure (see Ref. 16). This threshold has been evolved with time to be more conservative (see differences between Ref. 15 and Ref. 14).

c) Other metrics, e.g. clearly visible broken plies or hole (impact penetrating the whole thickness in thin laminates)



Figure 5. Breakage/delamination of front and back plies after impact (different locations)

d) Length of disbonds

This metric is need for dealing with disbond damages at bonded interfaces as joints between stringers and skins/webs in stiffened panels. A disbond of a stiffener of 100 mm length was shown reliably detectable by General Visual Inspection in Ref. 13. A disbond of this size is unlikely to be caused by a low velocity impact, thus frequently treated as a conservative upper bound of the BVID, if not directly as a VID.

A typical value of the energy cut-off applicable for BVID determination, for parts not specially exposed to severe impacts, of common use within Airbus, is **35J**. A detailed discussion on this topic can be found in Ref. 4, Ref. 5 and Ref. 3 / CMH-17 - Vol.3 - §12.9.1.

2.2 THE VID THRESHOLD

The VID (Visible Impact Damage) defines the upper limit of the damage that can be reliably detected during scheduled or directed field inspection and is not extremely improbable to occur (with probability of occurrence above $10^{-9}/\text{FH}$, the same of Ultimate Load event).

Maximum reliably detectable VID damages (threshold below consideration of the damage as obvious) are:

- Penetrating holes
- Sufficiently large disbonds or cracks

The maximum impact energy which needs to be considered, which is recognized as an extremely improbable event, since public references (see following figure, extract of Ref. 5), is **90J**. This figure may limit the severity of the VID at some locations (medium to high thickness).

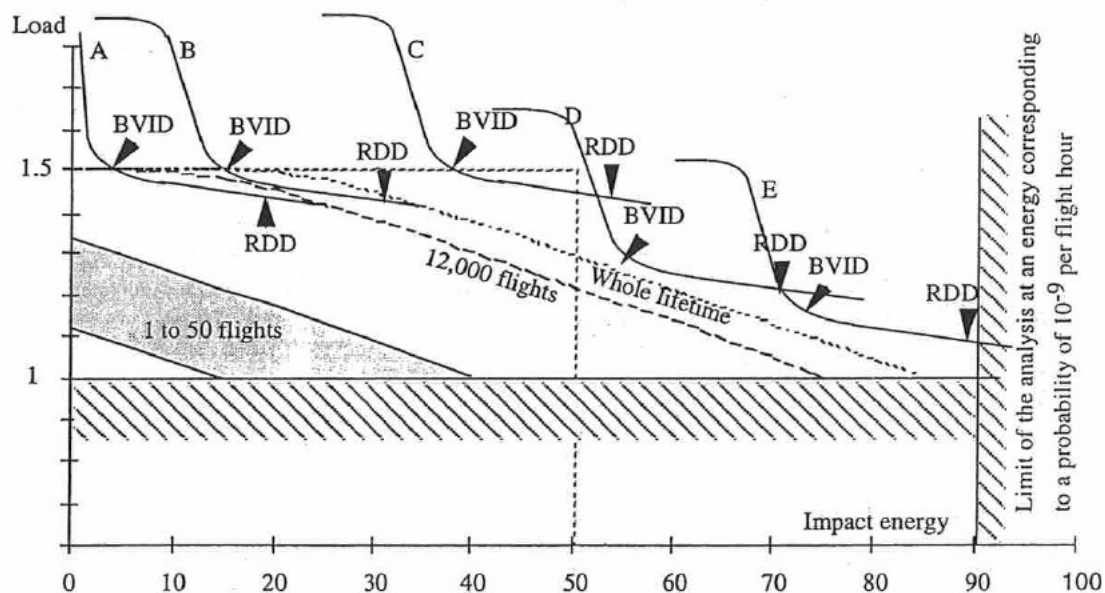


Figure 6. A typical recognized energy cut-off at the extremely improbable event is 90J

Remark: Details out of the scope of this report at current released issue:

- ✓ The 90J cut-off for the VID could be challenged by adequate probabilistic analysis (and inspection procedures and intervals). Advice on this topic in list of references.
- ✓ For aircraft locations especially exposed to impact damage threats (as cargo doors or root sections of fuselage mounted horizontal tail planes) the extremely improbable impact may be up to 140J

3 DAMAGE THREATS DURING...

3.1 ... MAINTENANCE

Sensitivity to low velocity impact damages, such as those from tool drops during fabrication or maintenance operations, is at the forefront of the damage tolerance evaluation for composite structures. Provided the impactor is blunt enough (*), drastic compression strength reductions may occur before damage becomes visually detectable.

(*) As the 16mm spherical impactor used within Airbus, are recognized the de-facto accepted standard causing “the most critical and least detectable damage” (Ref. 2)

Field surveys about accidental damages are available and yet mentioned by J. Rouchon (Ref. 5 and Ref. 6), including DOT/FAA reports Ref. 7 and Ref. 8. **Airbus surveys????** Cut-off energies limiting the severities of the BVID and VID, discussed in previous sections of this document, are derived from data reported on those surveys.

Those references recognize that the probabilities of occurrence of the impact events of some severity were evaluated for the whole aircraft, and anyway it is recommended to apply the cut-off values systematically to all the parts (instead of challenging those values by e.g. using ratios of exposed areas vs total aircraft surface). At the same time, it is implicitly recognized that considering simultaneously multiple BVID (or VID) damages at close distances would not be realistic.

Realistic impact scenarios for BVID determination are expected even more restrictive than those described for manufacturing in following section as, once the structure is assembled, there are less possible kinds of impact trajectories that may cause undetectable damage.

3.2 ... MANUFACTURING

3.2.1 Realistic low velocity impacts in composite stiffened panels

Similar types of damage threats, as well as thresholds for damage detectability (visual inspection) and probability of occurrence, which were discussed for maintenance activities, are valid for manufacturing.

The following figures provide a justification rational for constraining the potentially unlimited impact event types, which may occur, and thus enable an affordable approach for structural substantiation.

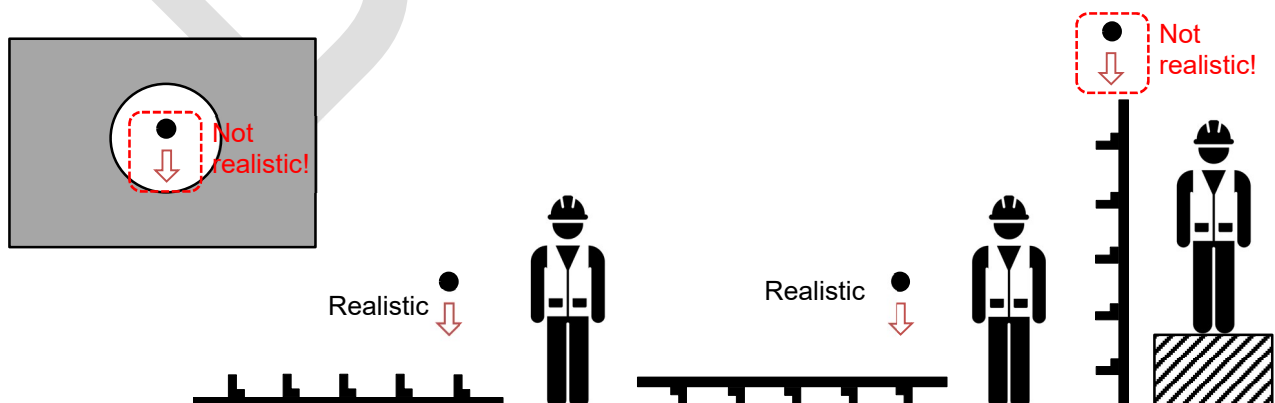


Figure 7. Tool drop realistic and unrealistic scenarios during composite panels manufacturing

Realistic:

- Transverse impact in panels (internal and external surface, inner areas, near panel or holes edges)
- Edge impacts in tee stringer webs, transverse impacts in stringers feet and in Z or hat stringers heads

Non-realistic:

- Edge impacts in holes edges
- Edge impacts in panel edges
- Transverse impacts in stringers webs



Figure 8. Realistic impacts in composite panels ("skins")

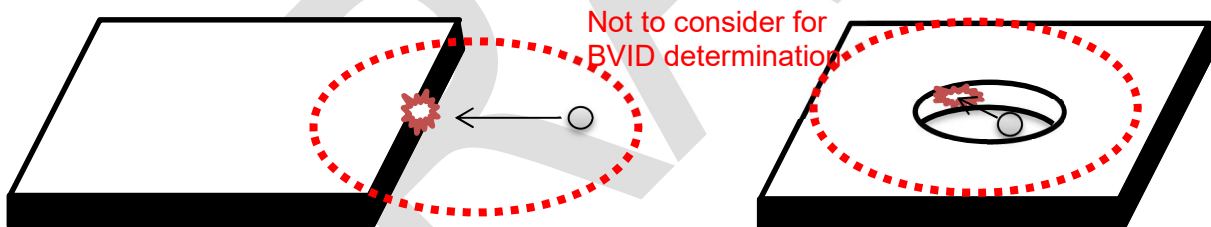


Figure 9. Un-realistic impact damages in composite panels ("skins")

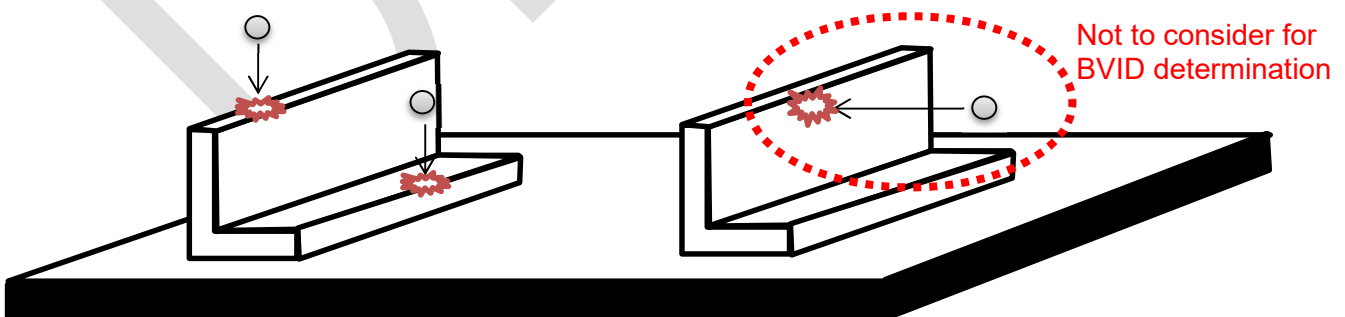


Figure 10. Realistic and un-realistic impacts in stringers of composite panels

3.2.2 Large damages

Anomalous events, as vehicle collisions, jacking errors, overload conditions..., which may create severe damage, are expected to be recognized by operations personnel, and activate and process including analysis of the event, inspection, eventually repair or scrap the part as required. Associate damages shall me then classified into Category 5 damages.

3.3 ... GROUND & FLIGHT OPERATIONS

3.3.1 Hail impact

Ground hail

“Composite aircraft structure should be resistant to typical hail that may be encountered while the aircraft is on the ground to minimize the amount of repair required after a hailstorm. The ground hail threat is most severe on upward facing surfaces, but is also of concern for vertical surfaces. In practice, ground hail resistance is typically achieved through the use of one or more minimum gage requirements” (CMH-17 Vol. 3, §12.5.2.5).

Note that the most severe hail impact damage which may occur shall be one of 14-15J of kinetic energy, this figure associated to a 90% of cumulative probability of avoiding the event per Table 12.5.2.5 of CMH-17 Vol. 3 (data in that table is based on hail records collected by the US National Oceanographic and Atmospheric Administration between 1955 and 2006). Most of damage tolerant composite parts in an aircraft shall retain ultimate load capability after an impact damage of 15J, as their BVID threshold will be typically well above that figure (note minimum cut-off energy required by authorities as acceptable limit for the BVID is 35J).

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TABLE 12.5.2.5 Summary for the cumulative US hail distribution from 1955 to 2006.

Diameter		Volume		Mass		Terminal Velocity		Kinetic Energy		Cumulative Probability of Occurrence
in	cm	in ³	cm ³	lb	g	ft/s	m/s	in-lb	J	
0.25	0.64	0.01	0.13	0.0003	0.12	32.29	9.84	0.05	0.01	
0.50	1.27	0.07	1.07	0.0021	0.97	45.66	13.92	0.83	0.09	
0.75	1.91	0.22	3.62	0.0072	3.26	55.93	17.05	4.21	0.47	21.7%
1.00	2.54	0.52	8.58	0.0170	7.72	64.58	19.68	13.31	1.50	48.1%
1.20	3.05	0.90	14.83	0.0294	13.34	70.74	21.56	27.61	3.10	64.7%
1.25	3.18	1.02	16.76	0.0333	15.08	72.20	22.01	32.51	3.65	68.1%
1.50	3.81	1.77	28.96	0.0575	26.06	79.09	24.11	67.41	7.57	81.4%
1.70	4.32	2.57	42.15	0.0837	37.94	84.20	25.66	111.21	12.49	88.3%
1.75	4.45	2.81	45.98	0.0913	41.39	85.43	26.04	124.88	14.03	89.6%
1.76	4.47	2.85	46.78	0.0928	42.10	85.67	26.11	127.76	14.35	89.9%
2.00	5.08	4.19	68.64	0.1362	61.78	91.33	27.84	213.04	23.93	94.4%
2.04	5.18	4.45	72.84	0.1446	65.56	92.24	28.11	230.60	25.91	95.0%
2.25	5.72	5.96	97.73	0.1940	87.96	96.87	29.52	341.24	38.34	97.1%
2.40	6.10	7.24	118.61	0.2354	106.75	100.05	30.49	441.75	49.63	98.1%
2.50	6.35	8.18	134.07	0.2661	120.66	102.11	31.12	520.11	58.43	98.5%
2.75	6.99	10.89	178.44	0.3541	160.60	107.09	32.64	761.49	85.55	99.3%
2.76	7.01	11.01	180.40	0.3580	162.36	107.29	32.70	772.63	86.80	99.3%
3.00	7.62	14.14	231.67	0.4597	208.50	111.85	34.09	1078.49	121.16	99.7%
3.25	8.26	17.97	294.54	0.5845	265.09	116.42	35.48	1485.48	166.88	99.8%
3.50	8.89	22.45	367.88	0.7301	331.09	120.82	36.82	1998.04	224.47	99.9%
3.75	9.53	27.61	452.47	0.8979	407.22	125.06	38.12	2633.04	295.80	100.0%
4.00	10.16	33.51	549.13	1.0898	494.22	129.16	39.37	3408.57	382.93	100.0%
4.25	10.80	40.19	658.67	1.3071	592.80	133.13	40.58	4343.98	488.02	100.0%
4.50	11.43	47.71	781.87	1.5516	703.68	136.99	41.75	5459.87	613.38	100.0%

Figure 11. Ground hail statistics (*) in CMH-17 Vol. 3, section 12.5.2.5

(*) The "Cumulative Probability" shown refers to the probability of avoiding such an event

In any case, a common design practice shall be to design with a minimum skin gage in which ground hail has no significant effect on structural strength, and thus no need of (frequent) repair. Specifically, the recommended design criterion would be to size for having no damage (no dent, no delamination...) for hail impacts up to 2J (*) (associated to a 50% cumulative probability in abovementioned table). Alternatively, designing with configurations equivalent to others installed in existing aircrafts and a vast service experience without relevant concerns regarding hail damage, should be considered also as a valid design (without further specific demonstrations).

(*) Note that the BVID cut-off energy shall be typically well above this value of 2J

Multiple hail impacts

The 2J impact selected as design limit for no-damage is associated to a hail impact of diameter of 1 inch. Further references with surveys data about hail stones events report impact densities of hail stones of maximum to 1 inch diameter from 11 impacts (average) up to 400 impacts (maximum) within one square feet. Translated to a square of 100mm side these figures come to 1.2 impacts (average) up to 44 impacts (maximum). From the last maximum number of 44 impacts only 2.4 would be of 2J severity (22/9).

The 100mm reference length is of interest as it is the width of the typical coupons used for determining residual strengths including BVID damages.

A reasonable strategy for substantiating no damage along the aircraft service life, at least for the abovementioned probability of 50%, shall be to demonstrate that 3 impacts of 2J in an area of 100mm x 100mm will not cause any significant damage in the composite material (verified by visual and NDT).

Table 3-3. Number of Hailstones Per Hailfall of a Given Diameter or Greater Per ft²

Diameter	Number of Hailstones per ft ²	
	Average	Maximum
0.5	9.22	290.
0.75	1.52	88.
1.0	0.45	22.
1.5	0.064	2.1
2.0	0.019	0.45
3.0	0.003	0.05
4.0	0.0007	0.01

Figure 12. Hail stone density statistic per Ref. 12

The requirements set for ground hail stone damages, as described in previous paragraphs, should not be a design driver for composite parts configured as solid laminates (damage thresholds well above 2J impacts), but would need some dedicated demonstration for less robust parts (as sandwich panels with thin faces).

In-flight hail

“Some structure, such as leading edges, windscreens, and forward facing fuselage structure, can experience in-flight hail impacts.... For example, the two primary threat conditions to be addressed for in-flight hail impact events on military aircraft are the 90% hail threat for retention of design ultimate load capability, and the 99% extreme hail threat for continued safe flight and landing” (CMH-17 Vol. 3, §12.5.2.6).

TABLE 12.5.2.6 Typical air speeds and hail diameters for medium- and long-haul aircraft

Cumulative Probability of Occurrence, %	Air Vehicle Velocity, KTAS (ft/sec)	Hail Diameter, inches
90	375 (633)	1.76
95	409 (690)	2.04
99	455 (768)	2.75

The first condition (impact energy 784 J), for lift boxes, can be covered by the bird strike impact substantiation (typ. dedicated test) if it is shown that the primary structure (front spar) is not damaged after the impact. For control surfaces, typically configured as sandwich panels, it shall be expected that sizing shall retain ultimate load capability with a full-penetrating damage (<2 inches).

The second condition can be covered for the leading edge by demonstrating compliance with bird strike impact requirements.

3.3.2 Bird strike

Reserved for future revisions of this chapter.

3.3.3 Lightning strike

Reserved for future revisions of this chapter.

DRAFT

4 APPENDICES

ABBREVIATIONS AND SYMBOLS

<i>Symbol</i>	<i>Units</i>	<i>Definition</i>
BVID	-	Barely Visible Impact Damage. It is the damage in a composite laminate consequence of a low velocity impact which may occur at the threshold of (visible) detectability (limit of "Category 1" damages)
DET/DVI	-	Detailed Visual Inspection. Intensive visual examination requiring access, proximity, adequate lighting (grazing light) and any necessary inspection aids (such as mirror or hand lens)
DSG	-	Design Service Goal
DV	-	Design Value, i.e. the value for a mechanical property of a structural material to be used in stress analyses
FRP	-	Fibre Reinforced Polymer
GVI	-	General Visual Inspection, which may require some access (removal of fairings, access panels) and use of ladders or workstand to gain proximity
II	-	Inspection Interval
LL	-	Limit Load
HTP	-	Horizontal Tail Plane
NDT	-	Non Destructive Technique
VID	-	Visible Impact Damage. Threshold defining the maximum damage that needs to be considered as a "Category 2" damage.
VTP	-	Vertical Tail Plane
UL	-	Ultimate Load

REFERENCES

	<i>Document Reference</i>	<i>Issue</i>	<i>Title</i>
Ref. 1	[EASA] CS-25	Amd.21	Certification Specifications and Acceptable Means of Compliance for Large Aeroplanes
Ref. 2	[EASA] AMC 20-29	07/2010	Composite Aircraft Structure (or FAA Advisory Circular AC No: 20-107B)
Ref. 3	[SAE] CMH-17	07/12	COMPOSITE MATERIALS HANDBOOK
Ref. 4	[J. Rouchon]	1990	CERTIFICATION OF LARGE AIRPLANE COMPOSITE STRUCTURES, RECENT PROGRESS AND NEW TRENDS IN COMPLIANCE PHILOSOPHY
Ref. 5	[J. Rouchon]	1999	EFFECTS OF LOW VELOCITY IMPACT DAMAGE ON PRIMARY COMPOSITE

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AIRCRAFT STRUCTURES. THE CERTIFICATION ISSUE.

Ref. 6	[J. Rouchon]	2009	Fatigue and Damage Tolerance Evaluation of Structures: The Composite Materials Response
Ref. 7	DOT/FAA/AR-96/111	1997	Advanced certification methodology for composite structures
Ref. 8	DOT/FAA/AR-95/17	1997	Development of probabilistic design methodology for composite structures
Ref. 9	[Airbus] AM2396.1	Is. A	Structure Analysis Categorization for Definition, Justification and Certification Phases
Ref. 10	[Airbus] MM00PR0510641	Is. 2	Composite Materials Impact damage policy (A400M)
Ref. 11	[Aerospatiale]	Is. 7	A330/A340 AIRFRAME DISCUSSION PAPER. SUBJET: COMPOSITE – IMPACT DAMAGE.
Ref. 12	[Jet Propulsion Laboratory]	1977	ENVIRONMENTAL HAIL MODEL FOR ASSESSING RISK TO SOLAR COLLECTORS
Ref. 13	IR 065/02	2002	FAN-COWL OMEGA SPECIMEN FOR TESTING PURPOSES
Ref. 14	[Airbus] ME0905804	Is. 1.0	PRELIMINARY UPDATE TO BVID REQUIREMENTS FOR EDGE IMPACTS ON A350 XWB COMPOSITE STRUCTURES
Ref. 15	[Airbus] RP0404309	Is. 1	Composite BVID criterion
Ref. 16	[Airbus] V57RP1039013	Is. 4.0	Edge Impact Damage Tolerance Method for A350
Ref. 17	[Airbus] AITM1-0076	Is. 1	Airbus Test Method Fibre. Reinforced Plastics. Determination of compression and tension strength after edge impact

SOFTWARE

Program	Version	Description

Table 1. Software programs referenced in this chapter

COMPOSITE DAMAGE THREATS

RECORD OF REVISIONS

Revision Date	Authors	Change Reason	Pages/Sub- chapters affected
0.4 23/01/2024	G. Baños	First issue	All pages

Table 2. Record of Revisions: authors, change reasons and sections/pages affected

APPROVAL SIGNATURES TABLE

Dpt. \ Site	Getafe	Manching
Structural Analysis	Javier Gómez-Escalonilla DD/MM/2024	Holger Hickethier DD/MM/2024
Stress Methods & Optimization	G. Baños 23/01/2024	-

Table 3. Approval signatures table for last revision of this chapter